Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/16/2023
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	11/04/2021	**End Date:**	11/03/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/03/2023 per NSSC information (Ed., 11/3/22)		
Key Personnel Changes/Previous PI:	This is a transfer of funding from Dr. Elison Blancaflor (formerly at the Noble Research institute) to the University of Wisconsin-Madison. The postdoc working on the project (Dr. Sabrina Chin) moved from the Noble Research Institute to the University of Wisconsin-Madison to continue with the work and is now a CoI on the project.		
COI Name (Institution):	Chin, Sabrina Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	80NSSC22K0029		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: This project is a continuation with Principal Investigator (PI) Dr. Simon Gilroy at the University of Wisconsin of "The Use of Microgravity Simulators for a Mechanistic Understanding of Cytoskeletal-Mediated Regulation of Root Growth," grant 80NSSC18K1462. The previous PI was Dr. Elison Blancaflor, Noble Research Institute, from July 2018-March 2021. Plants will be a major component of advanced life support systems to enable NASA to realize its vision of long duration space exploration that will take humans to Mars and beyond. However, plants can only be effectively used for such systems if we have a deeper mechanistic understanding about how their development is regulated by the spaceflight environment. Recent "omics" studies have shown that plants reprogram their genetic circuitry when grown in the near weightless conditions of space. Genes involved in cell wall remodeling and oxidative stress are major gene categories that are differentially regulated by microgravity. Further, it has been shown that microgravity in space and simulated microgravity using clinostats or random positioning machines elicit roughly similar root skewing responses in the model plant Arabidopsis thaliana. Root skewing was more pronounced in mutants to a root-expressed vegetative actin isoform. Consistent with the enhanced root skewing of actin mutants in microgravity is the observation that other plant species such as maize (Zea mays L.) have stronger root directional growth responses on a clinostat when chemicals such as latrunculin B disrupt their actin cytoskeleton. Based on these observations, we will test the hypothesis that the actin cytoskeleton is a key player in dictating root growth directionality in microgravity and simulated microgravity by regulating the process of autotropic organ straightening. Moreover, we hypothesize that a set of cell wall cross-linking (i.e., class III peroxidases) and putative actin-regulatory proteins (i.e., HLB1 and MIN7/BEN1) mediate root hair elongation and directional root growth, respectively, under microgravity and simulated microgravity. Our hypotheses will be addressed through three specific aims including the analysis of cell wall composition/distribution in wild-type Arabidopsis roots, and detailed characterization of selected Arabidopsis cell wall and actin mutants subjected to microgravity simulation on a slow rotating clinostat and random position machine (aim 1 and 3). We will also use microgravity simulators to address the hypothesis that the plant hormone brassinolide regulates directional root growth in maize through its modulation of actin organization and dynamics (aim 2). Our research plan builds on successes of previous NASA-funded ground and spaceflight results, and outlines a comprehensive set of experiments including cell wall immunocytochemistry, computer-based root growth analyses, actin quantification, and genetics. The proposed work aligns with Plant Biology (PB) Element of the Space Biology Science Plan. Results obtained from plant experiments on microgravity simulators will address guiding questions linked to the decadal survey recommendations including the impact of gravity on plant growth, development and metabolism (PB-1), and molecular mechanisms underlying how plants sense and react to gravity (PB-5). We expect that the proposed research will provide us with basic knowledge that will guide the development of plant cultivars better adapted to the microgravity environment of spaceflight, an outcome that will prove beneficial for the design of plant habitats on future space colonies and for addressing great agricultural challenges on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal aims to gain a better mechanistic understanding of how plant roots function at the cellular and molecular levels. A primary focus of the proposed work is to ask how the microgravity environment modulates the directionality of root growth. Because roots are essential for the uptake of water and nutrients, knowledge gained from successful implementation of the proposed work will be crucial in guiding pre-breeding efforts aimed toward the development of crops with improved productivity through better and more efficient root systems. Thus, agriculture on Earth is one beneficial outcome of this project.
Task Progress:	This grant is a continuation of NASA grant 80NSSC18K1462, which was transferred from the Noble Research Institute to the University of Wisconsin (80NSSC21K0029) in November 2022. Research accomplishments since the grant was transferred to Wisconsin are outlined below. Remaining experiments are related to aims 1 and 3. Experiments in aim 2 were completed at the Noble Research Institute. Specific Aim 1. Immunocytochemistry of selected cell wall components in roots of Arabidopsis thaliana seedlings grown on a slowly rotating clinostat and random positioning machine (RPM). Status: In progress. For this reporting period, research has prioritized completing and validating the glycomics and cell wall immunocytochemistry of seedlings from the Advanced Plant Experiments / APEX-03 experiments. Glycomics uses a panel of several hundred antibodies against non-cellulosic cell wall polymers to define the cell wall composition on the plant. Analysis has been completed of the cell wall changes elicited by spaceflight, as part of the APEX-03 experiment, using this glycomics profiling approach. This analysis provided a survey-level monitoring of wall composition that highlighted xyloglucans as a major wall polymer being modified in spaceflight. Monoclonal antibodies that recognized xyloglucan, xylan, and arabinogalactan epitopes were then further analyzed by immunocytochemistry to ask where within the plant these changes occurred. This approach showed that, e.g., the xylan backbone epitope recognized by antibody M140 was more intensely labeled in the xylem of space-grown roots than in ground-controls. Similarly, the M2-xyloglucan recognized by antibody M58 was more intense in the periphery of the spaceflight-grown roots. These results suggest that the spaceflight environment may trigger root secondary cell wall formation and modulate xyloglucan-related polymer production. This research used the high alkaline extracts from RNA purification columns as the input samples. RNA purification usually separates and then discards these cell wall fractions to purify the nucleic acids for subsequent sequencing. Therefore, this glycomics analysis from the column waste of RNA purification may add value to the transcriptomics analyses already used for plant spaceflight experiments. However, as this represents a new approach to cell wall sample preparation, validation is essential and was performed using identical extraction procedures on mutants defective in known aspects of cell wall chemistry: xxt1/xxt2 and mur3. The xxt1/xxt2 mutant has no xyloglucan, while the mur3-3 mutant contains cell wall xyloglucans that lack a galactose-fucose side chain. Root tissues of the mutants displayed the expected reduction in fucosylated xygloglucan, indicating this analysis of RNA extraction column residue is likely a robust approach to the monitoring of cell wall composition in spaceflight samples, even those already processed for RNA extraction. These findings have been published in NpjMicroavity in March of 2023. [Ed. Note: See Bibliography.] Specific Aim 2. Characterize autotropic root straightening in maize roots in response to brassinolide (BL) - and latrunculin B (LatB) - induced actin disruption. Status: Completed at the Noble Research Institute.

	Specific Aim 3. Evaluate root growth of selected Arabidopsis thaliana actin single and double mutants, and peroxidase mutants on microgravity simulators. Status: In progress. The responses of Col (wild type), spi5 (hlb2), act2-3, act7-4, act8-2, act2-3, act8-2 plants have been analyzed for their responses to growth on microgravity simulators. Actins 2, 7, and 8 are the major vegetative actins in Arabidposis and are thought to generate specific subsets of the microfilament arrays of the plant cell. SPIRIG was identified as an actin-related gene in a screen for hypersensitivity to the actin disrupting compound latrunculin. Actin 7 mutants showed low germination and severely delayed growth and development, and so were excluded from further analyses due to this pleiotropic phenotype. Wild-type plants showed significant root skewing, when grown on the clinostat, when compared to 1 x g controls. The act2 and act8 single mutants also skewed, whereas the act2-3/act8-2 double and spi5 mutant did not suggesting a potential effect on actin-dependent growth responses entrained to gravity.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Nakashima J, Pattathil S, Avci U, Chin S, Sparks JA, Hahn MG, Gilroy S, Blancaflor EB. "Glycome profiling and immunohistochemistry uncover changes in cell walls of Arabidopsis thaliana roots during spaceflight." npjMicrogravity. 2023 Aug 22;9(1):68. https://doi.org10.1038/s41526-023-00312-0 ; Pubmed PMID: 37608048; Pubmed Central PMCID:10444889 , Aug-2023
Articles in Peer-reviewed Journals	Bakshi A, Swanson SJ, Gilroy S. "A touchy subject: Ca2+ signaling during leaf movements in Mimosa." Cell Calcium. 2023 Mar;110:102695. https://doi.org/10.1016/j.ceca.2023.102695 ; PMID: 36669253 , Mar-2023